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PROGRAMME	MSc. Cybersecurity
Module Instructor:	Swati Dongre
Module/Subject Title	Advanced Programming Techniques
Module Code	B9CY108
Github Link	Github Link

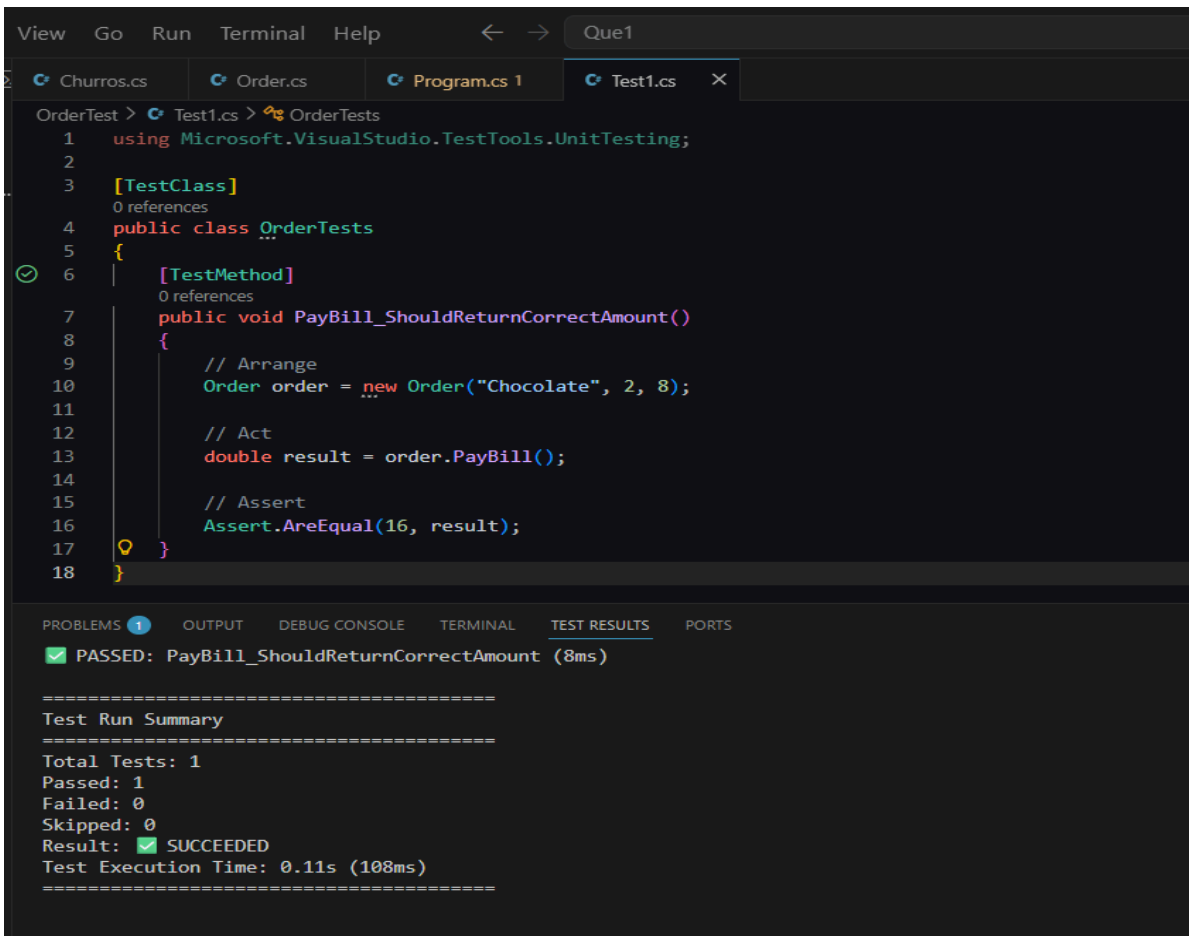
1. Introduction

The use of Python and C# programming expertise is the focus of this assignment. Building simple programs to comprehend key ideas like object-oriented programming, data structures, networking and data handling was the primary goal.

2. Task 1 – Churros Ordering System (C#)

For this assignment, I developed a Simple console application for the "Delicious Churros" food stand. The application enables the user to manage the order process, place orders, and determine the overall cost. In this program, I used two classes.

In the first “Churros” class item's name and price are stored. In the other “Order” class the quantity, order number, and total bill, are stored. Moreover, this made it easier for me to use object-oriented programming to correctly structure the code. I also used the FIFO rule to handle orders which mean the first order will be fulfilled first. In the end, I added a small unit test using MSTest to check if the bill calculation is correct.

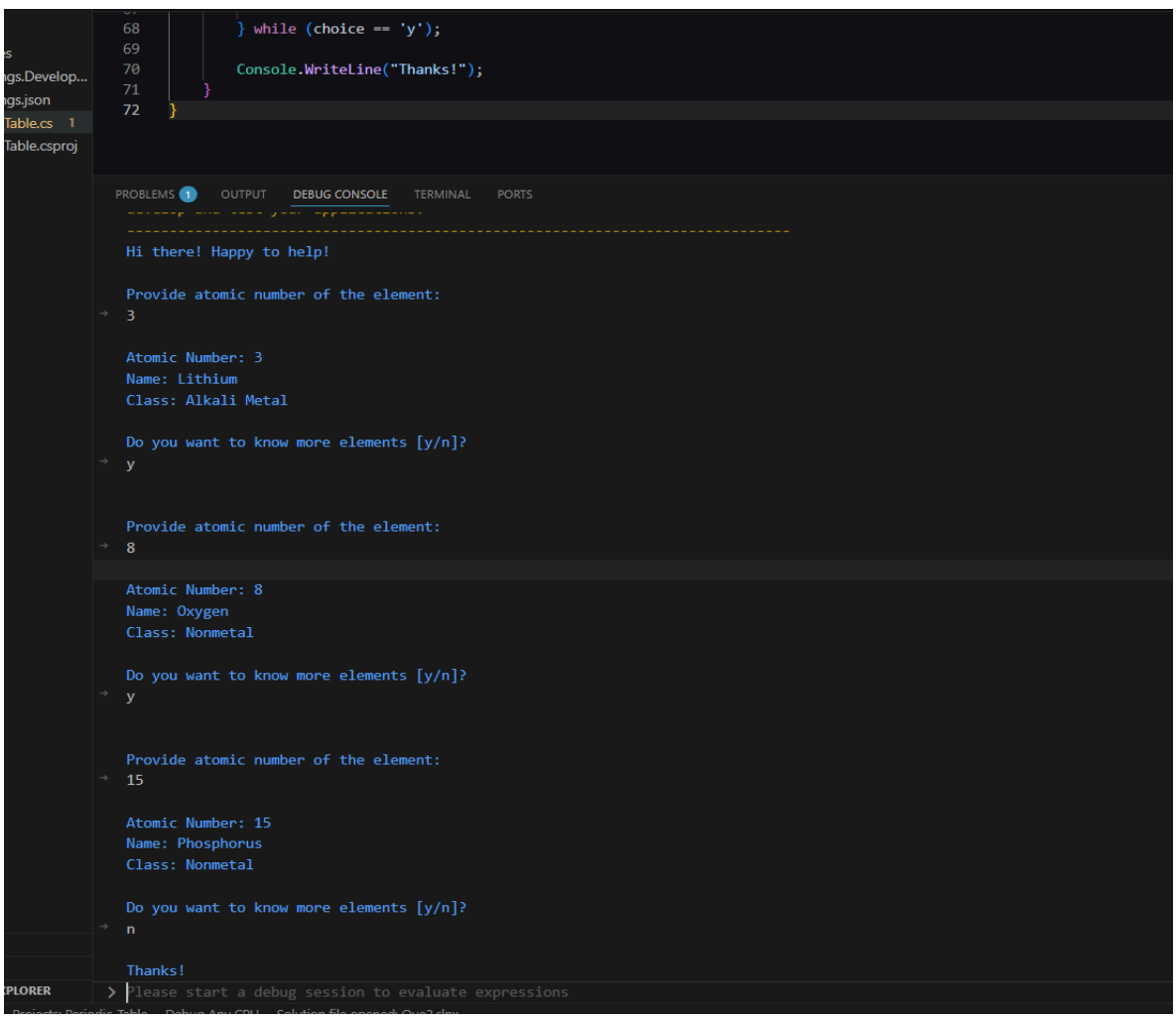


For task 1, Unit test for the Churros Ordering System was successful with the PayBill_ShouldReturnCorrectAmount test passing as shown in the console. This result confirms that C# logic for calculating the total bill based on the product price and quantity is functioning accurately within the Order class.

3. Task 2 – Periodic Table Application (C#)

I created a program for this assignment that saves the first thirty elements of the periodic table. I stored the data in a dictionary, where the element details are the value and the atomic number is the key.

This program will display the element name and type when the user enters an atomic number. For the save side, I added input checking so the program does not crash if the user enters wrong data. In order to allow the user to search again and again without restarting the application, a loop is also in this task. I learned how to use a dictionary to store and retrieve data rapidly thanks to this assignment.



In the task 2, The implementation of the Periodic Table Application successfully demonstrates the practical use of dictionaries in C# for efficient data retrieval, By mapping atomic numbers as keys to element details.

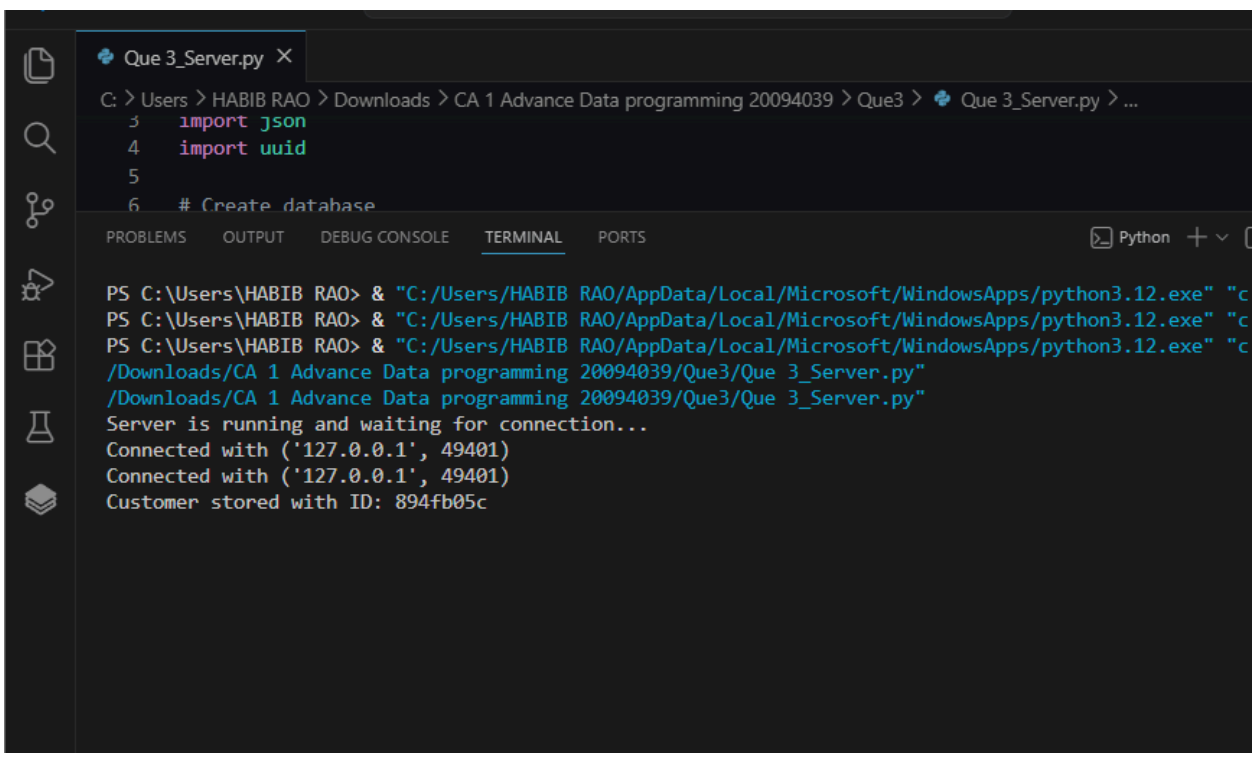
4. Task 3 – Client-Server Application (Python)

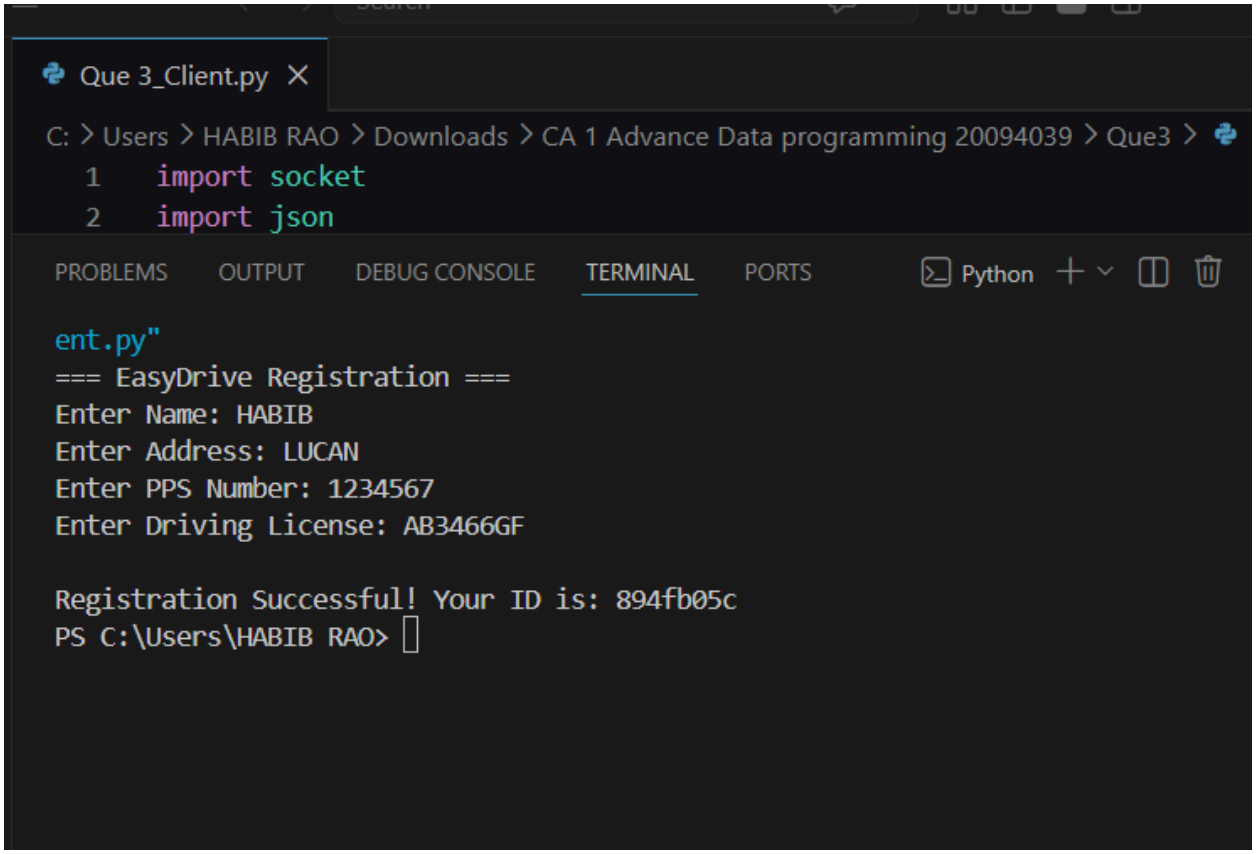
In this task, I developed a simple client-server system for a car rental service called “EasyDrive”. The client collects user details like Name, Address, PPS number, and Driving Licence, and sends them to the server.

The communication between client and server was done using socket programming. The data was sent in JSON format so it is easy to read and process. On the server side, the data was received and stored temporarily using variables instead of a database.

After receiving the data, the server creates a unique ID and sends it back to the client as confirmation. I also used basic cryptography to make the data transfer more secure.

This task helped me understand how two programs can communicate with each other.





For task 3, The Client-Server Application for “Easydrive” effectively showcases how two independent programs can communicate using socket programming. By using the JSON format for data transfer, it ensures that the information remains structured and easy for the server to parse upon receipt. The terminal results confirm that the server is successfully listening for connections and processing registrations in real-time.

5. Task 4 – Web Scraping and CSV Handling (Python)

In the last task, I worked on getting data from a website. I used the “requests” library to open the webpage and “BeautifulSoup” to extract information from it.

The program collected book names, prices, and ratings. This data was then saved into a CSV file.The file was readable and shown on the screen once it had been saved. This task helped me learn how to collect data from websites and store it in files.

```
    + CategoryInfo          : ObjectNotFound: (black:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\HABIB RAO> & "C:/Users/HABIB RAO/AppData/Local/Microsoft/WindowsApps/python3.12.exe" "c:/Users/HABIB RAO/Download
a programming 20094039/Que4/WebScriping.py"
Data saved to books.csv

--- Book Data ---
['Book Name', 'Price', 'Rating']
["It's Only the Himalayas", 'Â£45.17', 'Two']
['Full Moon over Noahâ\x80\x99s Ark: An Odyssey to Mount Ararat and Beyond', 'Â£49.43', 'Four']
['See America: A Celebration of Our National Parks & Treasured Sites', 'Â£48.87', 'Three']
['Vagabonding: An Uncommon Guide to the Art of Long-Term World Travel', 'Â£36.94', 'Two']
['Under the Tuscan Sun', 'Â£37.33', 'Three']
['A Summer In Europe', 'Â£44.34', 'Two']
['The Great Railway Bazaar', 'Â£30.54', 'One']
['A Year in Provence (Provence #1)', 'Â£56.88', 'Four']
['The Road to Little Dribbling: Adventures of an American in Britain (Notes From a Small Island #2)', 'Â£23.21', 'One']
['Neither Here nor There: Travels in Europe', 'Â£38.95', 'Three']
['1,000 Places to See Before You Die', 'Â£26.08', 'Five']
PS C:\Users\HABIB RAO>
```

For Task 4, The project successfully demonstrates how to Python for automated data collection through Web Scraping. By utilizing the requests library to fetch webpage content and BeautifulSoup to parse it, This showed a solid ability to extract specific data such as book title, Prices and rating from a complex HTML structure.

6. Tools and Technologies

For this assignment, I used C# in Visual Studio and Python in Visual Studio Code. I also used Python libraries like socket, requests, and BeautifulSoup. GitHub was used to save and manage my project files.

7. Conclusion

This assignments helped me improve my programming skills in both C# and Python. I learned how to use classes, data structures, and networking. I also learned how to work with real data. Overall, it was a good learning experience and helped me understand how to build simple applications.

8. References

- **C# (Microsoft)**
- [C# Documentation (Microsoft Learn)]

- **Python (Official documentation)**
- https://docs.python.org/3/

- **Socket programming (Python)**
- https://docs.python.org/3/library/socket.html

- **Requests library**
- https://requests.readthedocs.io/

- **BeautifulSoup**
- BeautifulSoup Documentation

- **Books Toscrape**
- https://books.toscrape.com